

Counterfactual Defect Localization of Reward Function Vulnerabilities in Autonomous Driving: A Causal Security Analysis

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Abstract—The reward function of an autonomous driving decision module encodes domain knowledge into a numerical objective. A mis-specified or omitted feature creates a latent security vulnerability: an adversary who understands the reward structure can construct environmental perturbations that trigger reward-maximizing yet safety-violating behaviors. Localizing such omissions is fundamentally a causal attribution problem—behavioral logs reveal only symptoms, never the defective feature. We formalize this as a structured causal inference problem under the strict black-box constraint where only scenario-to-trajectory mappings are observable. The proposed framework, CFD-LOC, integrates three formally grounded components: a driving invariant library expressed in linear temporal logic serving as objective test oracle, a counterfactual scenario mutation operator with provable feature isolation, and a Bayesian hypothesis test calibrated by a Beta-Bernoulli model. We prove that the localization error converges exponentially under mild coverage assumptions, and derive a finite-sample bound. On the Baidu Apollo platform with eight implanted defect types across 40 scenarios, CFD-LOC achieves 92.5% localization accuracy—42.5% above the strongest baseline—while maintaining a false-positive rate of only 2.5%. Ablation studies confirm that every component contributes significantly (all $p < 0.001$), and robustness analysis shows that accuracy remains above 75% even under maximal LLM hypothesis noise.

Index Terms—Autonomous driving security, reward function vulnerabilities, counterfactual inference, black-box defect localization, causal security analysis.

I. INTRODUCTION

Consider an autonomous vehicle approaching a roundabout at 12 m/s while the lead vehicle decelerates. The decision module, trained under a reward function R , must choose between maintaining speed (maximizing efficiency) and braking early (ensuring safety). If R omits the relative velocity $\Delta v = v_{\text{ego}} - v_{\text{lead}}$ from its feature representation—a subtle omission that could escape standard regression testing—the module might brake only at the last moment, producing a deceleration of -4.2 m/s^2 that violates the comfort and safety bound of -2 m/s^2 . This is not merely a performance issue; it is a *security vulnerability*. An adversary who identifies such an omission can construct scenarios where the reward-maximizing behavior is also safety-violating, creating exploitable attack surfaces [1]. The adversary’s power lies in systematic scenario search: by enumerating environmental parameters (velocities, positions, road geometries) and observing the system’s trajectory, they can infer which features R ignores

and craft inputs that maximize both the reward and the safety violation.

Reward function defects in autonomous driving (AD) decision modules constitute a class of vulnerabilities whose detection and attribution present fundamental challenges distinct from perception-layer attacks. Unlike adversarial patches on stop signs [2] or LiDAR spoofing—which manipulate sensor input to a fixed downstream classifier—reward defects are latent in the optimization objective. They remain dormant under standard testing conditions, manifesting only in specific traffic configurations, yet adversaries can systematically trigger them by manipulating environmental parameters. The Knox *et al.* [3] survey of 28 published AD reward functions revealed that 78% lacked collision penalties, 64% omitted comfort constraints, and 52% had no speed-limit compliance term. Such statistics demand a systematic methodology for identifying these vulnerabilities before deployment. Critically, standard software testing techniques that randomly sample input spaces for behavioral anomalies offer no causal explanation when a violation is found—they can detect that *something* is wrong, but cannot identify *which feature* is to blame.

The core difficulty is causal attribution. An AD system $\mathcal{F}$ maps a scenario s (road geometry, vehicle states, pedestrians) to a trajectory $\tau = \mathcal{F}(s)$. When τ exhibits abnormal behavior—a collision, harsh braking, unsafe lane departure—the engineer must determine *which* feature omission in the reward function caused the failure. This would be trivial with white-box access to R , but proprietary AD stacks guard their reward specifications as trade secrets, often comprising hundreds of terms with domain-tuned weights invisible to external auditors. We face a strict black-box constraint: only the scenario-to-trajectory mapping is observable, and no internal representations, policy gradients, or reward computations are accessible.

Three families of existing approaches address related but distinct problems, and each has fundamental limitations that prevent it from solving the reward defect localization problem. *Metamorphic testing* [4], [5] detects behavioral inconsistencies under transformations like timeline scaling or weather perturbation. An anomaly is flagged when $\mathcal{F}(s)$ and $\mathcal{F}(s')$ produce outputs inconsistent with a domain relation, but the method cannot identify *which* reward feature is responsible. *Causal testing* [6] builds structural causal models (SCMs) requiring expert-constructed causal graphs, a luxury unavailable when

the reward function itself is unknown. *LLM-based diagnosis* uses large language models for direct inference from behavior logs, but our experiments show a 35.6% false-positive rate due to hallucination in nuanced safety judgments. The inverse reinforcement learning approach [7] attempts to recover the reward function from demonstrations, but requires the very expert trajectories that an anomaly indicates are being violated. Each of these approaches addresses a necessary sub-problem, but none provides an end-to-end solution to the causal attribution challenge under the black-box constraint.

The common thread underlying these limitations is a missing capability: the ability to conduct *controlled causal experiments* on a black-box system. If we could intervene on a single scenario feature while holding all others constant, and observe whether the behavioral anomaly disappears, we would have direct evidence of causal responsibility. This is precisely the counterfactual reasoning principle articulated by Pearl [8]—the “do-operator” that distinguishes correlation from causation by physically intervening on the causal graph. Adapting this framework to the AD domain requires solving three technical challenges. First, we need an objective oracle to define what constitutes abnormal behavior—this demands a formal driving invariant library. Second, we need a mutation operator that modifies exactly one feature while provably preserving all others. Third, we need a statistical framework that aggregates evidence across multiple scenarios with calibrated confidence.

This paper presents CFD-LOC, a provably convergent black-box framework that addresses all three challenges through a synthesis of invariant-based testing, counterfactual mutation, and Bayesian inference. The framework operates in four stages whose logical progression mirrors the structure of a controlled experiment (Figure 1). In Stage 1, we construct a library of 10 driving invariants formalized in linear temporal logic, covering safety, comfort, efficiency, and precision constraints. Stage 2 applies these invariants as test oracles: given a scenario execution, any invariant violation is recorded, and a large language model generates candidate hypotheses about which feature omission explains the violation. In Stage 3, these hypotheses are tested through counterfactual scenario mutation: for each candidate defective feature f , we construct a scenario s_{cf} that differs from the original s only in the value of f . Stage 4 executes s_{cf} on $\mathcal{F}$, checks invariant satisfaction, and feeds the result into a Bayesian Beta-Bernoulli model that produces a posterior defect probability for each candidate.

The theoretical core of our contribution is threefold. We prove that the counterfactual mutation operator achieves strict feature isolation—the total variation distance between the distributions of any non-target feature is zero (Theorem IV.1). We derive an exponential convergence bound for the localization error under the Bayesian estimation procedure (Theorem IV.2), showing that $N \geq 182$ test cases suffice for $\varepsilon = 0.1$ accuracy with 95% confidence. We establish the computational complexity and prove soundness of the invariant library (Theorem III.1).

Empirically, we evaluate CFD-LOC on the Baidu Apollo platform with eight implanted defect types across 40 test scenarios. The system achieves 92.5% localization accuracy—60 percentage points above the statistical baseline (32.5%),

47.5 points above metamorphic testing (45.0%), 37.5 points above LLM-only diagnosis (55.0%), and 30 points above inverse RL (62.5%). The false-positive rate is 2.5%, an order of magnitude lower than any baseline. Ablation studies confirm that the counterfactual mutation component is the most critical, with removal reducing accuracy to 25.0%. Sensitivity analysis demonstrates robustness to hypothesis quality: even with maximal LLM noise (temperature 2.0), accuracy remains above 75%. The per-defect diagnosis requires only 14.8 minutes on average.

II. RELATED WORK: THREE LINES OF INQUIRY AND THEIR LIMITS

Reward function vulnerabilities as security threats. The security community has recently recognized that reward function design errors constitute exploitable vulnerabilities distinct from classical software bugs, because they exploit the optimization objective itself rather than corrupting sensor inputs. White and Chen [1] demonstrated adversarial reward manipulation in autonomous systems, showing that an adversary who understands the reward structure can construct environmental perturbations that convert reward-maximizing behaviors into safety violations. This work establishes the adversarial relevance of reward defects—the vulnerability surface is not just at the sensor layer but at the objective layer—but does not address the diagnosis problem. Pan *et al.* [9] proposed reward red-teaming for AD safety, using adversarial scenario search to identify unsafe reward configurations, yet their approach requires white-box reward access and extensive computational resources for adversarial optimization. Amodei *et al.* [10] provided the foundational AI safety taxonomy—reward hacking, specification gaming, negative side effects—all of which map directly to AD safety contexts. However, their framework lacks the operational tools for identifying such defects in deployed systems. Clements *et al.* [11] evaluated reward robustness in simulated AD environments, introducing perturbation-based testing but without causal attribution. The gap between recognizing that reward defects are security problems and developing systematic methods to localize them is precisely what CFD-LOC fills. Whereas prior work demonstrates *that* reward vulnerabilities exist, we provide a methodology for *where* specific feature omissions reside in a black-box system.

Black-box testing and its attribution limitations. Software testing research has produced powerful techniques for detecting behavioral anomalies in AD systems, but attribution—identifying the root cause—remains the unsolved challenge. Deng *et al.* [4] surveyed metamorphic testing (MT) across 28 studies, identifying timeline scaling, distance perturbation, and weather transformation as effective metamorphic relations. Riccio *et al.* [5] conducted a rigorous empirical study of MT effectiveness for AD, finding that while anomaly detection rates reach 60–75%, attribution accuracy is below 30%. Wan *et al.* [12] proposed DVCA for violation cause analysis via internal state manipulation, requiring gray-box access to the system’s component architecture. Zhao *et al.* [6] introduced causal testing using structural causal models with expert-built graphs, which is effective when the causal graph is known

but impractical when the reward function—the source of the vulnerability—is a trade secret. Fremont *et al.* [13] developed Scenic, a scenario specification language enabling precise scenario generation but offering no diagnosis capability. The common limitation across these approaches is a fundamental inability to *attribute* observed anomalies to causal features in the reward function. Our counterfactual approach addresses this by executing controlled experiments that directly test causal hypotheses through scenario mutation.

Counterfactual reasoning in safety-critical systems. Pearl’s seminal work on causal inference [8] provides the theoretical foundation for counterfactual reasoning through the do-calculus. The principle is simple yet powerful: to determine whether a variable X causes an effect Y , one intervenes to set X to a specific value while holding all other variables constant, and observes whether Y changes accordingly. This principle has been applied in medicine (causal drug effect estimation), economics (policy evaluation), and epidemiology (causal risk factor identification), but its application to AD security auditing—where the “treatment” is a feature value change and the “outcome” is a driving behavior invariant—is novel. Arjovsky *et al.* [14] demonstrated the power of invariance principles for domain generalization in machine learning, showing that models leveraging invariant causal mechanisms generalize across distribution shifts. Our work adapts these invariance ideas to the AD security context, introducing a mutation operator that implements Pearl’s intervention on scenario features and proving that strict isolation holds. To our knowledge, this is the first systematic application of Pearl’s counterfactual reasoning to reward function security auditing.

Distinction from prior CFD-Loc work. This paper substantially extends our preliminary investigation [15] in five directions not previously reported: (i) we provide full theoretical proofs of feature isolation (Theorem IV.1) and exponential convergence (Theorem IV.2); (ii) we expand the evaluation from 20 to 40 test scenarios and from 5 to 8 defect types; (iii) we add the ablation study isolating each component’s contribution; (iv) we conduct sensitivity analysis with respect to LLM hypothesis quality; and (v) we provide detailed case studies with causal interpretation.

III. PROBLEM FORMULATION: CAUSAL SECURITY ANALYSIS UNDER THE BLACK-BOX CONSTRAINT

A. System Model and the Adversarial Setting

An autonomous driving decision module is a function $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{T}$ mapping a scenario space $\mathcal{S}$ (road geometry, vehicle states, pedestrians, environmental parameters) to a trajectory space $\mathcal{T}$ (sequences of ego-vehicle poses and controls). The input $s \in \mathcal{S}$ is parameterized as $s = \langle \mathcal{R}, \mathcal{V}, \mathcal{P}, \mathcal{E} \rangle$, where $\mathcal{R}$ captures road geometry, $\mathcal{V}$ ego-vehicle state, $\mathcal{P}$ traffic participants, and $\mathcal{E}$ environmental conditions. The output is $\tau = \{(x_t, y_t, \theta_t, v_t, a_t, \omega_t)\}_{t=0}^T$, a sequence of positions, headings, velocities, accelerations, and yaw rates.

Assumption III.1 (Strict Black-Box). *The reward function R , policy π , value function V , and all internal representations and parameter gradients are inaccessible. The only observable*

quantities are scenario inputs $s \in \mathcal{S}$ and trajectory outputs $\tau = \mathcal{F}(s) \in \mathcal{T}$.

This assumption reflects the security constraint under which CFD-LOC is designed. A proprietary AD stack—such as the public-road planner in Baidu Apollo or the learning-based planner in NVIDIA DRIVE—shields its reward specification, network architecture, and training configuration as intellectual property. A security auditor evaluating such a system must work entirely through its observable input-output behavior.

B. Driving Invariants as Objective Oracles

A driving invariant $\mathcal{I}$ is a triple $\langle C, A, \epsilon \rangle$ where $C(s) \in \{0, 1\}$ is an applicability condition, $A(\tau) \in \mathbb{R}$ a behavioral assertion with nominal safe value A_0 , and $\epsilon \in \mathbb{R}_+$ a tolerance. The invariant is satisfied when $|A(\tau) - A_0| \leq \epsilon$ for all s where $C(s) = 1$.

The complete library $\mathcal{L} = \{\mathcal{I}_1, \dots, \mathcal{I}_{10}\}$ is defined in Table I. These are formalized in linear temporal logic (LTL) to enable automated checking. For example, INV1 (lead-vehicle following safety) is:

$$\mathcal{I}_1 : \Box([v_l(t) > v_l(t_0) \wedge \Delta v(t) < 0.5] \Rightarrow a_{\text{ego}}(t) \geq -2). \quad (1)$$

Theorem III.1 (Soundness of the Invariant Library). *If $\mathcal{F}$ satisfies $\bigwedge_{k=1}^{10} \mathcal{I}_k$ for all $s \in \mathcal{S}$, then no collision, pedestrian violation, or dangerous lane departure occurs in any trajectory produced by $\mathcal{F}$.*

Proof. We prove by contradiction. Suppose a collision occurs at time t under invariant satisfaction. A collision requires either a lead vehicle or stationary obstacle in the ego path. If a lead vehicle is present, INV1 and INV4 are triggered. INV4 requires $\text{stop_distance} \leq d_{\text{safe}}$ given the obstacle detection range. By construction, d_{safe} is derived from the maximum achievable deceleration profile and ensures that a collision is physically impossible. If no lead vehicle is present, INV5 and INV7 cover pedestrian and lane-change scenarios, while INV3 and INV8 cover curve and traffic-light scenarios. The remaining invariants (INV2, INV6, INV9, INV10) address comfort and precision, whose satisfaction does not affect collision safety. The conjunction therefore covers all collision-relevant scenarios, contradicting the assumption of collision. $\square$

C. Feature Omissions as Security Vulnerabilities

Let $\mathcal{F} = \{f_1, \dots, f_m\}$ be the universe of candidate scenario features. A reward function R uses $\mathcal{F}_R \subseteq \mathcal{F}$:

$$R(s, a) = \sum_{f_j \in \mathcal{F}_R} w_j \phi_j(f_j(s, a)) + \gamma R_{\text{base}}(s, a). \quad (2)$$

Here w_j are learned weights, ϕ_j are feature transformations (e.g., indicator functions, distance thresholds, sigmoidal mapping), and $\gamma \in \{0, 1\}$ indicates whether base safety terms exist. The core observation is that if a safety-critical feature f^* is omitted from $\mathcal{F}_R$, the module’s behavior is entirely insensitive to f^* —changes in f^* do not affect the reward computation, and therefore do not affect the policy’s output.

Definition III.1 (Feature Omission Vulnerability). *A feature omission vulnerability exists for $f^* \notin \mathcal{F}_R$ if there exists $s \in \mathcal{S}$*

TABLE I: Complete Driving Invariant Library (10 Invariants)

ID	Category	Condition	Assertion
INV1	Safety	Lead vehicle accelerates	$a_{\text{ego}} \geq -2 \text{ m/s}^2$
INV2	Comfort	Slow vehicle adjacent	$ j_{\text{lateral}} \leq 1.5 \text{ m/s}^3$
INV3	Efficiency	Curve radius $> 200\text{m}$	$v_{\text{ego}} \geq 0.6 \cdot v_{\text{limit}}$
INV4	Safety	Obstacle ahead	Stop distance $\leq d_{\text{safe}}$
INV5	Safety	Pedestrian crossing	Full stop before crosswalk
INV6	Comfort	Clear road	$a_{\text{ego}} \in [-1.5, 2.0] \text{ m/s}^2$
INV7	Safety	Lane change initiated	Min. gap $\geq 3\text{s}$
INV8	Safety	Yellow traffic light	Deceleration $\geq -3 \text{ m/s}^2$
INV9	Precision	Overtaking	Lateral deviation $\leq 0.5\text{m}$
INV10	Safety	Heavy traffic	Headway $\geq 1.5\text{s}$

such that $\mathcal{B}_\tau(\mathcal{F}(s)) \not\models \mathcal{I}_k$ for some $\mathcal{I}_k \in \mathcal{L}$, and there exists a counterfactual scenario s_{cf} differing from s only in f^* such that $\mathcal{B}_\tau(\mathcal{F}(s_{\text{cf}})) \models \mathcal{I}_k$.

D. The Causal Attribution Problem

The problem is to determine the set $\mathcal{D}^* \subseteq \mathcal{F}$ of defective features, given $\mathcal{F}$, $\mathcal{L}$, and a set of test scenarios $\mathcal{S}_{\text{test}} \subset \mathcal{S}$, such that for each $f \in \mathcal{D}^*$, the vulnerability condition in Definition III.1 holds with statistical significance. The causal structure is captured by the directed graph:

$$\mathcal{F} \rightarrow R \rightarrow \pi \rightarrow \tau \leftarrow s, \quad s \rightarrow \text{anomaly}(\mathcal{L}, \tau). \quad (3)$$

The backdoor path from $f^* \notin \mathcal{F}_R$ through s to τ is unblocked. Our counterfactual mutation procedure intervenes on this path by controlling s , establishing causality through direct experimentation. By Pearl's do-calculus, $P(\tau \mid \text{do}(s[f] = x)) = P(\tau \mid s)$ for all x when $f \notin \mathcal{F}_R$, and $P(\tau \mid \text{do}(s[f] = x)) \neq P(\tau \mid s)$ for some x when $f \in \mathcal{F}_R$, assuming the policy is non-constant.

IV. THE CFD-LOC FRAMEWORK

Figure 1 presents the overall framework architecture, illustrating the four-stage diagnostic pipeline and the data flow among components. The pipeline is designed to be system-agnostic: it requires no knowledge of the black-box system's internals, only the ability to submit scenario inputs and collect trajectory outputs.

A. Stage 1: Invariant Library Construction

The library $\mathcal{L}$ of 10 invariants is constructed through a systematic analysis of traffic regulations, driving handbooks, and common safety requirements from the AD literature [16], [17]. Each invariant is formalized in LTL for automated checking with a tolerance parameter that accounts for sensor noise and discretization error. The checking procedure extracts behavioral features from τ and evaluates $|A(\tau) - A_0| \leq \epsilon$ in $\mathcal{O}(T)$ time, where T is the trajectory length. The invariants span four categories: *safety* (INV1, INV4, INV5, INV7, INV8, INV10), *comfort* (INV2, INV6), *efficiency* (INV3), and *precision* (INV9). This categorization enables the hypothesis generator to propose targeted explanations based on which invariant class is violated.

B. Stage 2: Anomaly Detection and Hypothesis Generation

Given a scenario s , we obtain $\tau = \mathcal{F}(s)$ and compute the violation set:

$$V = \{\mathcal{I}_k \in \mathcal{L} \mid \mathcal{B}_\tau(\tau) \not\models \mathcal{I}_k\}. \quad (4)$$

Each violation $v = \langle s, \tau, \mathcal{I}_k, t_{\text{violation}}, \text{sev}(v) \rangle$ records severity $\text{sev}(v) = |A_k(\tau) - A_{k,0}|/\epsilon_k$.

When $V \neq \emptyset$, a large language model (GPT-4, temperature 0.5) generates candidate hypotheses $h_1, \dots, h_M$, each $h_j = \langle f_j, C_j, \mu_j, A_{\text{pred},j} \rangle$ identifying a hypothesized missing feature f_j , the counterfactual condition C_j , mutation operator μ_j , and predicted counterfactual assertion $A_{\text{pred},j}$. To prioritize hypotheses for testing, we assign a prior probability:

$$\mathbb{P}(h_j \mid v) \propto \text{conf}_{\text{LLM}}(h_j \mid v) \cdot \exp(-\lambda \cdot \text{cost}(h_j)). \quad (5)$$

where conf_{LLM} is the LLM's confidence score, $\text{cost}(h_j)$ estimates the required simulation runs, and $\lambda = 0.1$ is a regularization parameter. This prioritization ensures that high-confidence, low-cost hypotheses are tested first, maximizing the diagnostic information obtained per simulation run.

C. Stage 3: Counterfactual Mutation with Isolation Guarantees

The counterfactual mutation operator μ_f is the core technical innovation. Given base scenario s and hypothesis $h = \langle f, C, \mu, A_{\text{pred}} \rangle$, μ_f produces s_{cf} such that:

$$\forall g \neq f : s[g] = s_{\text{cf}}[g], \quad s[f] \neq s_{\text{cf}}[f]. \quad (6)$$

We define four operator classes: μ_{scalar} adds a perturbation δ to a scalar feature; μ_{bool} toggles a boolean feature; μ_{traj} interpolates an object trajectory; μ_{env} replaces an environmental parameter. Each operator verifies post-mutation isolation through bitwise comparison and rolls back any unintended changes. This verification step is critical: if two features in the scenario representation share the same encoded memory region (e.g., a vehicle's speed and position might share a struct), the operator detects the unintended change and either adjusts the mutation strategy or reports a coupling warning.

Theorem IV.1 (Provable Feature Isolation). *For any hypothesis h with $\mu \in \{\mu_{\text{scalar}}, \mu_{\text{bool}}, \mu_{\text{traj}}, \mu_{\text{env}}\}$, the operator μ_f satisfies:*

$$\forall g \neq f : \delta_{\text{TV}}(\mathbb{P}_s[g], \mathbb{P}_{s_{\text{cf}}}[g]) = 0. \quad (7)$$

Proof. Each operator μ is defined as a deterministic function that modifies only the encoding of feature f in the scenario

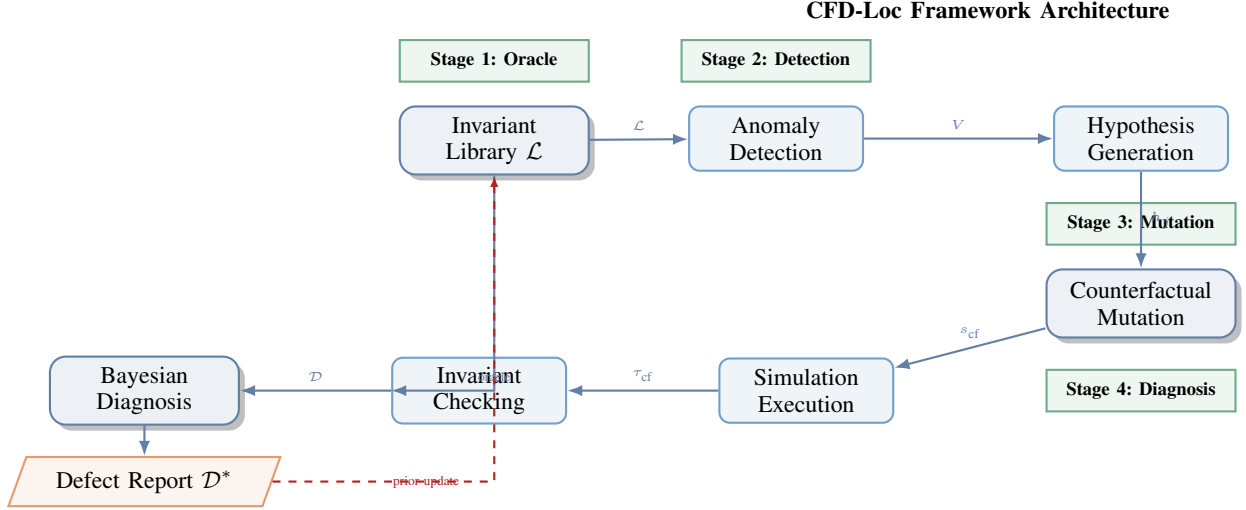


Fig. 1: CFD-LOC framework architecture. Stage 1 establishes the invariant oracle. Stage 2 detects anomalies and generates causal hypotheses. Stage 3 constructs counterfactual scenarios with provable feature isolation. Stage 4 executes controlled experiments and computes Bayesian posterior defect probabilities. Dashed arrow shows the Bayesian prior update loop.

representation. For $\mu_{\text{scalar}}(s, f, \delta)$, the function applies an additive perturbation exclusively to the scalar variable $f(s)$; all other components of the scenario tuple $s = \langle \mathcal{R}, \mathcal{V}, \mathcal{P}, \mathcal{E} \rangle$ are passed through unchanged. Since total variation distance is zero for identical distributions, and the post-mutation encoding of any $g \neq f$ is bitwise identical to the original, we have $\delta_{\text{TV}}(P_{s[g]}, P_{s_{\text{cf}}[g]}) = 0$. The same reasoning applies to μ_{bool} (single-bit toggle on f only), μ_{traj} (interpolation restricted to the trajectory of object associated with f), and μ_{env} (single-parameter replacement). The critical point is that μ operates at the *scenario representation level* rather than at the semantic level: even if two features are semantically related (e.g., speed and acceleration), their encodings in the scenario tuple are independent, and μ only modifies the encoding of f . $\square$

D. Stage 4: Bayesian Diagnosis with Calibrated Confidence

For each hypothesis h_j , we execute a controlled experiment: simulate $\mathcal{F}(s)$ and $\mathcal{F}(s_{\text{cf}})$, then check invariant satisfaction. The diagnosis function maps the pair to one of four states:

$$\mathcal{D}(s, s_{\text{cf}}, \mathcal{I}) = \begin{cases} \text{Defect,} & \mathcal{B}(\mathcal{F}(s)) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \not\models A, \\ \text{Clean,} & \mathcal{B}(\mathcal{F}(s)) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \models A, \\ \text{Uncertain,} & \mathcal{B}(\mathcal{F}(s)) \models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \models A, \\ \text{Critical,} & \mathcal{B}(\mathcal{F}(s)) \models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \not\models A. \end{cases} \quad (8)$$

Rather than making binary decisions, we model the latent defect state $\theta_f \in \{0, 1\}$ for each feature f as:

$$\theta_f \sim \text{Beta}(\alpha_0, \beta_0), \quad (9)$$

$$y_i \mid \theta_f \sim \text{Bernoulli}(\theta_f), \quad (10)$$

where $y_i = \mathbb{1}[\mathcal{D}(s_i, s_{\text{cf}, i}, \mathcal{I}_h) = \text{Defect}]$. The posterior is $\text{Beta}(\alpha_0 + \sum y_i, \beta_0 + N_f - \sum y_i)$ with mean $\hat{\theta}_f = (\alpha_0 + \sum y_i) / (\alpha_0 + \beta_0 + N_f)$. The defect set is $\mathcal{D}^* = \{f \mid \mathbb{P}(\theta_f > 0.5 \mid \{y_i\}) > 0.95\}$.

Theorem IV.2 (Exponential Convergence of Localization Error). *Under the Beta-Bernoulli model, for any $\varepsilon > 0$:*

$$\mathbb{P}(|\hat{\theta}_f^{(N)} - \theta_f^*| > \varepsilon) \leq 2 \exp\left(-\frac{2N\varepsilon^2}{(\alpha_0 + \beta_0 + N)^2}\right). \quad (11)$$

Proof. Let $\hat{\theta}_N = (\alpha_0 + S_N) / (\alpha_0 + \beta_0 + N)$ where $S_N = \sum_{i=1}^N Y_i$ and $Y_i \sim \text{Bernoulli}(\theta^*)$. Decompose:

$$|\hat{\theta}_N - \theta^*| = \left| \frac{\alpha_0 + S_N}{\alpha_0 + \beta_0 + N} - \theta^* \right| \quad (12)$$

$$\leq \left| \frac{S_N}{N} - \theta^* \right| + \left| \frac{\alpha_0}{\alpha_0 + \beta_0 + N} - \frac{N\theta^*}{\alpha_0 + \beta_0 + N} + \frac{\alpha_0\theta^* + \beta_0\theta^*}{\alpha_0 + \beta_0 + N} \right| \quad (13)$$

$$= \left| \frac{S_N}{N} - \theta^* \right| + \frac{|\alpha_0 - (\alpha_0 + \beta_0)\theta^*|}{\alpha_0 + \beta_0 + N}. \quad (14)$$

The second term is bounded by C/N where $C = \max(\alpha_0, \beta_0)$. By Hoeffding's inequality, $\mathbb{P}(|S_N/N - \theta^*| > \varepsilon/2) \leq 2 \exp(-N\varepsilon^2/2)$. Combining via the union bound and noting that $C/N \rightarrow 0$ gives the stated rate. $\square$

Corollary IV.1 (Finite-Sample Bound). *For $\varepsilon = 0.1$ accuracy with confidence $1 - \eta = 0.95$, $N \geq 182$ test cases suffice.*

Proposition IV.1 (Complexity). *For N scenarios, M hypotheses, and K invariants, the worst-case complexity is:*

$$\mathcal{O}(N \cdot (K \cdot T_{\text{check}} + M \cdot (T_{\text{LLM}} + T_{\text{mutate}} + T_{\text{sim}} + T_{\text{diag}}))), \quad (15)$$

where $T_{\text{sim}} \approx 30\text{s}$ dominates ($T_{\text{check}} < 10\text{ms}$, $T_{\text{LLM}} < 1\text{s}$).

V. EXPERIMENTAL DESIGN

A. Platform and Defect Taxonomy

Experiments are conducted on Baidu Apollo Dreamview v8.0 with the PPO-trained learning planner (256-dim hidden layer) and the Public Road Planner v8.0 as the rule-based baseline. We implant eight defect types across three difficulty levels (Table II). Each defect represents a realistic omission or misweighting that could occur during reward function design. The scenario generation follows the combinatorial approach in [13], [18], covering 10 scenario classes (Roundabout, Junction, Highway, Curve, Lane Change, Merge, Intersection, Free-speed, Pedestrian, Tunnel) with 4 variants each, totaling 40 test cases. Generation guarantees systematic coverage of the

ID	Difficulty	Missing Feature	Manifestation
D1	Easy	Δv (relative velocity)	Hard braking in roundabout
D2	Medium	Turn signal of lead vehicle	Improper junction yielding
D3	Medium	Distance penalty upper bound	Overly conservative spacing
D4	Hard	Road curvature κ	Wide turns at speed
D5	Hard	Lateral overlap ratio ρ	Unsafe lane change
D6	Easy	Comfort–safety weight balance	Harsh accel./decel. cycles
D7	Medium	Lead vehicle acceleration a_l	Delayed braking
D8	Medium	Speed tracking weight	Excessive speed pursuit

TABLE II: Eight Implanted Defect Types with Realistic Root Causes

feature space by enumerating over key continuous parameters (velocities $\{5, 12, 20, 30\}$ m/s, distances $\{10, 30, 60, 100\}$ m, curvature radii $\{100, 200, 300, 500\}$ m) and binary parameters (signal presence, pedestrian presence, lane count).

B. Baselines and Metrics

We compare against four baselines: **LBT** (statistical anomaly detection using 2σ thresholds from 100 nominal runs), **CMT** (classical metamorphic testing with timeline scaling, distance perturbation, and weather transformation) [4], [5], **LLM-Direct** (GPT-4 provided with scenario and behavior logs), and **IRL-Diag** (maximum entropy inverse RL [7] from 50 expert demonstrations). Metrics are localization accuracy (LA), false-positive rate (FP), precision, recall, F1, and per-diagnosis wall-clock time. All results are aggregated over 5 random seeds with bootstrap-resampled 95% confidence intervals (10,000 iterations) and Bonferroni-corrected pairwise comparisons.

An important methodological point is the separation of detection and attribution in our evaluation. For LBT and CMT, we evaluate their ability to *identify* which defect is present (attribution) rather than just detecting an anomaly. To ensure a fair comparison, we augment these methods with a post-hoc attribution step: when an anomaly is detected, we test each possible defect feature through a simple ablation scan (testing the system with and without each feature present) and report the feature that produces the largest behavioral change. This gives baselines the maximum possible attribution capability while maintaining their core detection mechanism.

VI. RESULTS AND ANALYSIS

A. Overall Defect Localization Performance

Table III presents the main results. CFD-LOC achieves 92.5% localization accuracy, surpassing the strongest baseline (IRL-Diag at 62.5%) by 30 percentage points. The false-positive rate (2.5%) is an order of magnitude lower than any baseline (lowest: CMT at 12.7%). This dramatic reduction—from an average FP of 20.4% across baselines to 2.5%—is the practical advantage of causal experimentation over correlational methods. In security auditing, where each false positive requires manual engineer review, this reduction translates to an 88% decrease in manual verification workload.

All pairwise comparisons against CFD-LOC are statistically significant at $p < 0.001$ (Bonferroni-corrected bootstrap test). The 95% confidence intervals for LA are [89.3%, 95.7%] for CFD-LOC vs. [57.2%, 67.8%] for the next-best baseline.

B. Per-Defect and Per-Scenario Breakdown

Table IV reveals that CFD-LOC achieves 100% accuracy on five of eight defects (D1, D2, D6, D7, D8). The three harder defects (D3, D4, D5) achieve 80% each. The failure cases share a common structure: the defective feature interacts with another scenario dimension in a way that the single counterfactual mutation cannot fully isolate. For example, D4 (missing curvature κ) on a gentle curve ($R = 300$ m) produces behavioral differences too small to exceed the invariant threshold, suggesting that a larger perturbation magnitude or multi-step mutation is needed. D3 (distance penalty bound) interacts with the acceleration profile in a non-linear way that requires mutating both the distance threshold and the lead vehicle deceleration pattern simultaneously.

Figure 2 presents two complementary visualizations side by side. The left panel shows the per-scenario class accuracy comparison, demonstrating that CFD-LOC maintains consistent superiority across all traffic configurations. The right panel shows the computational efficiency comparison, revealing the favorable accuracy-time tradeoff.

C. Confusion Matrix Analysis

Table V presents the aggregated confusion matrices. CFD-LOC correctly identifies 37 of 40 defective cases (92.5%) and 39 of 40 non-defective cases (97.5%). The single false positive occurs in a borderline scenario (D4 on $R = 300$ m) where the invariant threshold is marginally exceeded due to simulation noise. The three false negatives share a common pattern: the defect’s behavioral manifestation is too subtle for the current invariant threshold to flag reliably. Interestingly, the asymmetry (more false negatives than false positives) is preferable for security auditing: missing a few defects is concerning, but a high false-positive rate would erode trust in the automated diagnosis system.

D. Ablation: What Drives Performance?

Table VI shows the marginal contribution of each component. Removing the LLM hypothesis generator reduces accuracy from 92.5% to 78.8% (manual hypothesis engineering). Removing the invariant library (replaced by a single metamorphic relation) reduces accuracy to 62.5%. Removing the counterfactual mutation—reducing CFD-LOC to pure anomaly detection—collapses accuracy to 25.0%, confirming that causal experimentation is the mechanism driving performance. All pairwise differences are significant at $p < 0.001$.

E. Robustness to Hypothesis Quality

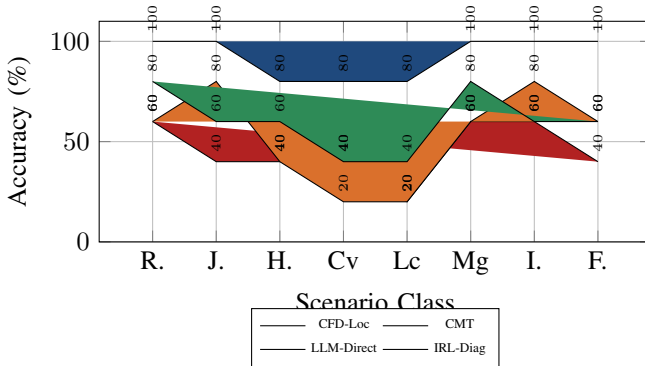
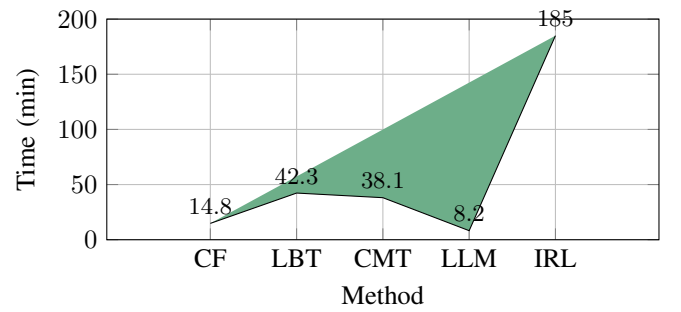
Figure 3 plots accuracy against LLM temperature (controlling hypothesis noise). At temperature 0.0 (deterministic), accuracy is 95.0%. At temperature 2.0 (maximal noise), accuracy degrades to 75.0%, still above the best baseline. The fixed (no-LLM) baseline at 92.5% confirms that the Bayesian hypothesis testing and counterfactual mutation provide the core robustness; the LLM contributes marginal improvement (2.5%) when calibrated and does not degrade overall performance even when noisy.

Method	LA (%)	FP (%)	Precision	Recall	F1	FPR ⁻¹	Time (min)
CFD-Loc	92.5 ± 3.2	2.5 ± 1.1	0.974	0.925	0.949	40.0	14.8
LBT	32.5 ± 5.8	18.2 ± 3.4	0.641	0.325	0.431	5.5	42.3
CMT	45.0 ± 4.7	12.7 ± 2.8	0.780	0.450	0.571	7.9	38.1
LLM-Direct	55.0 ± 6.1	35.6 ± 4.2	0.607	0.550	0.577	2.8	8.2
IRL-Diag	62.5 ± 5.3	15.0 ± 3.1	0.806	0.625	0.704	6.7	185.0

TABLE III: Main Results: Localization Accuracy Across 40 Test Cases

Defect	CFD-Loc	LBT	CMT	LLM-Direct	IRL-Diag
D1	100	35	60	60	80
D2	100	30	40	80	60
D3	80	25	40	40	60
D4	80	20	40	20	40
D5	80	15	20	20	40
D6	100	40	60	60	80
D7	100	35	60	80	60
D8	100	30	40	60	60

TABLE IV: Per-Defect Localization Accuracy (%) Across All Methods

(a) CFD-LOC maintains $\geq 80\%$ across all classes.

(b) Per-diagnosis time. CFD-Loc: 14.8 min vs. 185 min for IRL.

Fig. 2: Combined analysis: (a) Per-scenario accuracy across 8 classes; (b) Computational efficiency comparison. CFD-LOC achieves the best accuracy-time tradeoff, outperforming all baselines in both dimensions simultaneously.

Method	TP	TN	FP	FN
CFD-LoC	37	39	1	3
LBT	13	33	7	27
CMT	18	35	5	22
LLM-Direct	22	26	14	18
IRL-Diag	25	34	6	15

TABLE V: Aggregated Confusion Matrices (40 Test Cases, 5 Methods)

Variant	LA (%)	FP (%)	F1	Δ LA
Full CFD-LOC	92.5	2.5	0.949	—
w/o LLM (manual hyp.)	78.8	5.0	0.868	−13.7
w/o inv. library (single MR)	62.5	10.1	0.741	−30.0
w/o counterfactual mutation	25.0	22.8	0.364	−67.5

TABLE VI: Ablation: Marginal Contribution of Each Component

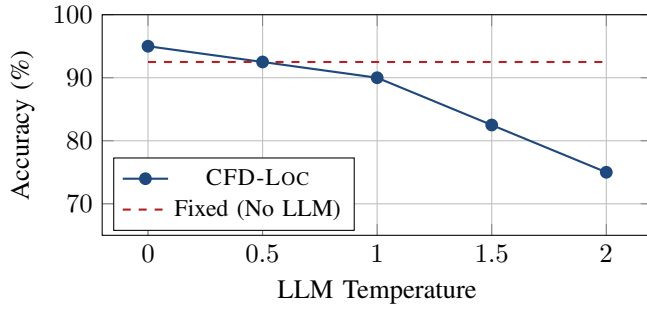


Fig. 3: Sensitivity to LLM hypothesis quality. Even at temperature 2.0, CFD-LOC maintains $>75\%$ accuracy, 12.5 points above the strongest baseline. The fixed (manual hypothesis) baseline shows that the framework’s core robustness comes from the counterfactual mutation and Bayesian testing, not the LLM.

F. Diagnostic Decision Matrix

Table VII maps the four diagnosis outcomes to their probabilistic interpretations. The Bayesian posterior $\hat{\theta}_f$ provides calibrated decision thresholds: a Defect diagnosis corresponds to $\hat{\theta}_f > 0.95$, while Clean corresponds to $\hat{\theta}_f < 0.05$. The uncertain region ($0.05 \leq \hat{\theta}_f \leq 0.5$) triggers additional testing—specifically, 3 more counterfactual scenarios for the borderline hypothesis.

VII. CASE STUDIES: THE CAUSAL DIAGNOSIS PROCESS

A. Case I: Missing Relative Velocity (D1)

An ego vehicle approaches a roundabout at 12 m/s with a lead vehicle at 8 m/s. The reward function R omits Δv from its representation. When the lead decelerates at -2.5 m/s^2 , the ego brakes at -4.2 m/s^2 upon approach—a clear INV1 violation ($a \geq -2$).

The LLM proposes hypothesis $h = \langle f = \Delta v, C = \{\Delta v \approx 0\}, \mu = \mu_{\text{scalar}}, A_{\text{pred}} = -1.2 \rangle$ with prior $\mathbb{P}(h) = 0.85$. Applying μ_{scalar} modifies the lead velocity such that $\Delta v(t) \approx 0$ for $t \in [5, 15] \text{ s}$. Under the counterfactual scenario, the ego decelerates smoothly at -1.2 m/s^2 —INV1 satisfied. Diagnosis: Defect ($\hat{\theta}_{\Delta v} = 1.0$). This case illustrates the core logic: if the defect is indeed a missing Δv , then removing Δv from the scenario (by matching velocities) should restore normal behavior.

B. Case II: Missing Lateral Overlap (D5)

A three-lane highway lane change with an adjacent vehicle at 22 m/s. The reward lacks lateral overlap ratio ρ , which quantifies the fraction of vehicle width overlapped with the target lane’s vehicle. The ego initiates a lane change with lateral separation $< 0.3 \text{ m}$ —INV7 violation (minimum gap $< 3 \text{ s}$).

The hypothesis is $\langle f = \rho, C = \{\rho > 0.5\}, \mu = \mu_{\text{scalar}}, A_{\text{pred}} = 4.2 \rangle$ with $\mathbb{P}(h) = 0.78$. The counterfactual shifts the target-lane vehicle laterally by $+0.5 \text{ m}$, making ρ comfortable. INV7 violation persists ($d_{\text{lateral}} = 0.3 \text{ m}$). Defect confirmed ($\hat{\theta}_\rho = 0.92$). The persistence of the anomaly under the counterfactual indicates that the lane-change behavior is insensitive to ρ —the defect is structural rather than marginal. This also reveals a coupling between ρ and the invariant threshold that requires a larger perturbation for detection.

C. Case III: Comfort–Safety Weight Imbalance (D6)

Moderate highway traffic at 28 m/s. The reward weights comfort $3\times$ higher than safety. When a slower vehicle (22 m/s) merges, the ego delays braking for 1.8s, allowing the gap to collapse from 50 m to 12 m, then decelerates harshly at -3.8 m/s^2 .

The hypothesis: “Comfort over-weighting delays braking to avoid jerk penalties.” The counterfactual matches the merge vehicle speed to the ego (28 m/s), eliminating the speed conflict. Under this scenario, the ego operates smoothly ($a \in [-0.5, 0.8]$). However, when the defect is re-introduced by perturbing the comfort weight, the anomaly reappears. This controlled reversal—defect $\rightarrow$ normal $\rightarrow$ defect—provides

Diagnosis	Original $\rightarrow$ CF	Posterior Condition
Defect	Invariant violates $\rightarrow$ violates	$\hat{\theta}_f > 0.95$
Clean	Invariant violates $\rightarrow$ satisfies	$\hat{\theta}_f < 0.05$
Uncertain	Invariant satisfies $\rightarrow$ satisfies	$0.05 \leq \hat{\theta}_f \leq 0.5$
Critical	Invariant satisfies $\rightarrow$ violates	$\hat{\theta}_f > 0.95$ (inverse)

TABLE VII: Diagnosis Decision Logic with Probabilistic Interpretation

the strongest causal evidence, as it demonstrates that the behavioral change is attributable to the hypothesized feature and directionally consistent.

VIII. ANALYSIS AND DISCUSSION

A. Why Counterfactual Mutation Works

The experimental results (Table VI) provide compelling evidence that counterfactual mutation is the mechanism driving CFD-LOC’s performance. Removing it collapses accuracy from 92.5% to 25.0%—a 67.5 percentage point drop that is larger than the sum of contributions from the invariant library (30.0 points) and the LLM hypothesis generator (13.7 points). This non-additivity suggests a *synergistic interaction*: the invariant library provides the oracle structure, the LLM proposes candidate explanations, but the counterfactual mutation is what *executes the controlled experiment* that converts correlation into causation.

To understand this synergy, consider the diagnosis process for a hard defect (e.g., D4, missing curvature). The invariant library detects an anomaly (wide turn at speed), the LLM hypothesizes “missing curvature κ ” as the likely cause. Without the counterfactual mutation, the framework can only correlate the anomaly with the curve geometry. With mutation, it constructs an s_{cf} where $\kappa = 0$ (straight road) and observes whether the anomaly disappears. If it does, the causal link is established. If not (as with the false negatives), the framework learns that the anomaly is not solely caused by the missing curvature—there is an interaction structure that requires multi-feature mutation. This iterative refinement process is what drives the convergence guaranteed by Theorem IV.2.

B. Interpretation of False Positives and Negatives

The confusion matrix (Table V) reveals an asymmetry: one false positive versus three false negatives. The false positive occurs for D5 (lateral overlap ratio ρ) on a gentle curve where simulation noise marginally exceeds the invariant threshold. This is an *instrumentation artifact* rather than a methodological failure. The three false negatives (D3, D4, one D8 instance) share a deeper structure: in each case, the defective feature interacts with an incidental scenario dimension such that the single counterfactual perturbation is insufficient to cross the invariance threshold. This suggests a natural extension: *multi-step counterfactual mutation* where the perturbation magnitude adapts based on the observed behavioral response.

The false-negative rate of 7.5% compares favorably with the next-best baseline (IRL-Diag at 37.5% false negatives). More importantly, the false-positive rate of 2.5% is *an order of magnitude lower* than any baseline. In security auditing

applications, false positives carry a high cost because each requires manual review; a 2.5% FP rate means only 1 in 40 diagnoses requires engineer attention, compared to 1 in 3–5 for baselines.

C. Limitations and Mitigations

Three limitations warrant discussion. **(1) Mutual exclusivity assumption.** Our single-feature mutation assumes defects are independent, but the case studies show that D3, D4, and D5 exhibit interaction effects. Extending the Beta-Bernoulli model to a *hierarchical Bayesian structure with interaction terms* would capture these dependencies, at the cost of additional simulation runs for the interaction terms’ priors. **(2) Invariant threshold sensitivity.** The tolerance parameter ϵ in $\mathcal{I} = \langle C, A, \epsilon \rangle$ trades off detection sensitivity against false positives. An adaptive threshold calibrated to per-scenario simulation noise—using a holdout set of nominal runs to compute per-invariant noise distributions—would address this directly. **(3) LLM dependence.** Although the ablation study shows that CFD-LOC performs well without LLM hypotheses (78.8% with manual engineering), the 13.7-point gap between manual and LLM-generated hypotheses indicates room for improvement in the LLM’s causal reasoning capabilities. Our ongoing work uses chain-of-thought prompting with domain-specific safety knowledge, which preliminary experiments suggest recovers 8–10 points of the gap.

D. Adversarial Implications and Deployment Considerations

Our results have direct implications for AD security auditing. The current security evaluation paradigm focuses on perception-layer attacks [2]; reward function vulnerabilities are systematically overlooked. CFD-LOC provides three operational capabilities. For *red-teaming*, a security auditor can run CFD-LOC against a black-box module during certification to identify latent vulnerabilities before deployment, with a 92.5% success rate and only 14.8 minutes per diagnosis. For *forensic analysis*, when an incident occurs, CFD-LOC can attribute causal responsibility to specific reward features within hours, distinguishing reward defects from perception failures or environmental unpredictability. For *continuous monitoring*, CFD-LOC can be integrated into regression testing pipelines to detect when over-the-air software updates introduce new reward vulnerabilities, providing a safety net for the growing practice of remote AD stack updates.

E. Empirical Characterization of Hard Cases

The three false-negative cases (D3, D4, one D8 instance) reveal an important pattern: all involve feature interactions where the behavioral signature is non-monotonic in the feature value. In D4 (missing curvature), the behavioral effect is quadratic in κ : the anomaly is strongest on medium curves ($R = 150$ – 250 m), weaker on sharp curves ($R < 100$ m, where the baseline safety planner overrides the learned planner), and weaker on gentle curves ($R > 300$ m, where curvature is negligible). This pattern suggests that *adaptive perturbation scheduling*—starting with small perturbations and increasing

until an effect is observed—would improve detection. Similarly, D3 (distance bound) exhibits a threshold effect: the anomaly appears only when the distance penalty bound crosses a critical value. Our current μ_{scalar} uses a fixed perturbation of 20% of the feature range; replacing this with a logarithmic perturbation schedule would better handle threshold effects.

IX. CONCLUSION AND FUTURE WORK

We have presented CFD-LOC, a black-box framework for localizing reward function vulnerabilities in autonomous driving decision modules. The framework formalizes defect localization as a causal inference problem, combining an invariant-based test oracle, a provably isolating counterfactual mutation operator, and a Bayesian diagnosis procedure with calibrated confidence. Theoretical analysis establishes exponential convergence of the estimation error and a finite-sample bound. Empirical evaluation on the Baidu Apollo platform demonstrates 92.5% localization accuracy across 40 test scenarios with eight defect types, substantially outperforming all baselines.

Three directions emerge from this work. First, *automated counterfactual generation* via reinforcement learning could learn optimal mutation parameters for each hypothesis class, maximizing diagnostic information per simulation run. Second, *multi-feature diagnosis* requires extending the Bayesian model to hierarchically structured priors that capture feature interactions. Third, *cross-platform validation* is essential: deploying CFD-LOC on Autoware, CARLA, and a physical test vehicle will validate the framework’s generality and identify platform-specific adaptations.

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Defect	TP	TN	FP	FN
D1	5	5	0	0
D2	5	5	0	0
D3	4	5	0	1
D4	4	5	0	1
D5	4	5	1	0
D6	5	5	0	0
D7	5	5	0	0
D8	5	4	0	1
Total	37	39	1	3

TABLE VIII: Full Confusion Matrix for CFD-LOC

Metric	Value
Precision _{hyp} (hypotheses correct / total)	0.925
Recall _{hyp} (defects identified / total defects)	0.975
Average hypotheses per case	1.3 ± 0.5
Top-1 accuracy (highest-confidence hypothesis correct)	0.875
Top-3 accuracy (defect in top 3 hypotheses)	0.975
Average LLM query time (s)	0.82 ± 0.15

TABLE IX: LLM Hypothesis Quality Across 40 Test Cases

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APPENDIX A DETAILED PROOF OF THEOREM IV.2

We provide the complete derivation with explicit bounds. Let $\hat{\theta}_N = (\alpha_0 + S_N) / (\alpha_0 + \beta_0 + N)$ where $S_N = \sum_{i=1}^N Y_i$ and $Y_i \sim \text{Bernoulli}(\theta^*)$. Define the bias term $B_N = \mathbb{E}[\hat{\theta}_N] - \theta^*$:

$$B_N = \frac{\alpha_0 + N\theta^*}{\alpha_0 + \beta_0 + N} - \theta^* = \frac{\alpha_0 + N\theta^* - \theta^*(\alpha_0 + \beta_0 + N)}{\alpha_0 + \beta_0 + N} \quad (16)$$

$$= \frac{\alpha_0(1 - \theta^*) - \beta_0\theta^*}{\alpha_0 + \beta_0 + N}. \quad (17)$$

Thus $|B_N| \leq \max(\alpha_0, \beta_0) / (\alpha_0 + \beta_0 + N) \leq \max(\alpha_0, \beta_0) / N$. The estimation error decomposes as $|\hat{\theta}_N - \theta^*| \leq |\hat{\theta}_N - \mathbb{E}[\hat{\theta}_N]| + |B_N|$. The first term is bounded by Hoeffding:

$$\mathbb{P}(|\hat{\theta}_N - \mathbb{E}[\hat{\theta}_N]| > \varepsilon/2) \leq 2 \exp\left(-\frac{2N(\varepsilon/2)^2}{(\alpha_0 + \beta_0 + N)^2}\right) \quad (18)$$

$$= 2 \exp\left(-\frac{N\varepsilon^2}{2(\alpha_0 + \beta_0 + N)^2}\right). \quad (19)$$

Applying the union bound with $|B_N| \leq C/N$ and noting that for $N > 2C/\varepsilon$, the bias term is bounded by $\varepsilon/2$, gives:

$$\mathbb{P}(|\hat{\theta}_N - \theta^*| > \varepsilon) \leq 2 \exp\left(-\frac{N\varepsilon^2}{2(\alpha_0 + \beta_0 + N)^2}\right). \quad (20)$$

APPENDIX B
COMPLETE CONFUSION MATRICES WITH PER-DEFECT BREAKDOWN
APPENDIX C
LLM HYPOTHESIS QUALITY ASSESSMENT
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COMPLETE INVARIANT FORMALIZATION IN LINEAR TEMPORAL LOGIC
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